

Hybrid privacy-aware semantic search: SVD-truncated document geometry and CKKS-encrypted query reranking under a restricted threat model

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Abstract

Dense embeddings power semantic search and Retrieval-Augmented Generation, yet a leaked vector database also leaks the text behind it, because embeddings can be inverted with high fidelity. The textbook defences are extreme—fully homomorphic search is sound but far too slow at million-document scale, while privacy noise degrades ranking before it protects. We study a middle path built on an asymmetry: the static document collection is protected *geometrically*—each vector is SVD-truncated onto a lower-dimensional subspace and then rotated by a secret orthogonal transform held only by the data owner—while the dynamic query is protected *cryptographically* under CKKS, so an honest-but-curious server never sees query values or similarity scores. The CKKS parameters are fixed by a small reproducible offline benchmark rather than by hand.

We prove a tight lower bound on the reconstruction error of any decoder confined to the protected subspace. On a one-million-document corpus with five encoders the protection preserves—and on the strongest encoders slightly improves—retrieval quality, a linear-denoiser effect, at sub-second latency, while an off-the-shelf inversion attack collapses to the noise floor. Stepping outside that regime we quantify the boundary: a known-plaintext attacker recovers the secret rotation by orthogonal Procrustes from about as many leaked pairs as the retained dimension, and the public product-quantization codes leak most neighbour structure.

The same asymmetric geometry doubles as a privacy-preserving semantic data-loss-prevention primitive for LLM firewalls: a server holding only the protected vectors detects whether a candidate matches a confidential reference corpus at near parity with a plaintext detector, degrading gracefully under text obfuscation. We state the limits plainly—query confidentiality is cryptographic, but document protection rests on SVD truncation and a secret rotation that form an empirical obfuscation layer, not a cryptographic primitive—and we delimit the threat model under which each claim holds.

Keywords: homomorphic encryption, CKKS, privacy-preserving information retrieval, semantic search, embedding inversion, non-adaptive attacker, restricted threat model, SVD truncation, product quantization, retrieval-augmented generation.

1 Introduction

The privacy of vector representations of text—“embeddings”—has become a first-class concern in production retrieval systems. State-of-the-art encoders (Sentence-BERT [1], multilingual-E5 [2], GTR [3], BGE-M3 [4]) push the per-document representation through a $\sim 10^2$ -dimensional bottleneck that nevertheless preserves enough information to reconstruct the original text with high BLEU. VEC2TEXT [7] reports BLEU $\approx 97.3\%$ on 32-token inputs when the attacker has query access to the encoder; GEIA [8] and TEIA [9] show that even snapshot attacks (an adversary with read access to the vector database) recover personally identifiable information at

high rates. For Retrieval-Augmented Generation pipelines whose vector stores typically contain customer support tickets, internal corporate documents and personal data, this rule of thumb—“a leak of the embedding is a leak of the text”—is now well documented [14].

The defensive landscape is bracketed by two extremes that, individually, do not satisfy industrial requirements. Differential privacy [18, 19] adds calibrated noise per-record but, in the naive coordinate-Gaussian formulation on retrieval-quality embeddings, destroys ranking quality long before the privacy budget reaches a level considered useful. Fully homomorphic encryption [23, 28] allows arbitrary computation on ciphertexts, but the ciphertext-ciphertext (ct-ct) regime is too expensive for top- k search across 10^6 documents.

Nature and positioning of this work. This is a *scientific study*, not a systems-engineering paper. SVD truncation, random orthogonal rotation, product quantisation and CKKS are individually well known; we do not claim a new system or a deployable product. The research questions are: **(RQ1)** what does projection provably remove—is there a formal lower bound on what any projection-restricted decoder can reconstruct? **(RQ2)** how do SVD truncation and an unknown secret rotation jointly reshape the privacy–accuracy–latency trade-off of semantic search at 10^6 -document scale? **(RQ3)** does data-driven SVD truncation behave as a destructive lossy step, or—as we find—as a linear *denoiser* for retrieval-trained encoders? **(RQ4)** does the same asymmetric geometry support a privacy-preserving *leak-detection* primitive—can a server flag when a candidate matches a protected reference corpus without storing readable embeddings of it, and at what fidelity relative to a plaintext detector? Accordingly, the secret rotation is studied as an empirical obfuscation phenomenon, not proposed as a cryptographic primitive, and the formal lemma is a projection bound, not an inversion-security theorem.

Object of study and scientific contribution. The object we analyse is an *asymmetric* hybrid construction (described below); the contribution is its theoretical and empirical characterisation under a credible—but explicitly restricted—threat model, not its engineering realisation:

- For the *static* side (the database), we apply a deterministic SVD truncation V_k with $k = d/2$, immediately followed by a Haar-uniform secret orthogonal rotation $R \in O(k)$. The truncation is chosen so that the proxy criterion $\sigma_{rec} \geq 0.10$ (Section 6) is met uniformly for all tested encoders; the rotation hides the orientation of the basis from any attacker that does not know R . We then publish a Product-Quantisation artefact (codebook + per-document PQ codes), trained *in the rotated space*, so that stage-1 candidate filtering can be performed locally by the client without sending the query to the server.
- For the *dynamic* side (the query), we apply CKKS encryption. The protocol is engineered so that the server-side reranking over the short-list of K_{cands} candidates only needs the ciphertext-plaintext (ct-pt) mode—the costly relinearisation step specific to ct-ct multiplication is avoided—bringing the cryptographic budget for one query well below one second.
- The CKKS parameters $(N_{poly}, \log_2 Q, d_{proj}, \Delta)$ are fixed by a small, reproducible offline micro-benchmark over the discrete parameter grid (Section 5): we keep the minimum-latency configuration that satisfies the conservative security tables and an accuracy tolerance. The selected configuration $(N_{poly} = 8192, [60, 40, 60], \Delta = 2^{40})$ yields a $\approx 1.7\times$ speed-up over the TenSEAL stock setting $[60, 40, 40, 60]$ at the same parameter-table security bound. This is a deterministic engineering choice over a published grid; we make no machine-learning claim.
- *A semantic DLP primitive.* We observe that the same construction doubles as a leak-detection index for LLM firewalls and AI gateways: the protected static side becomes a confidential reference corpus of sensitive or regulated documents, and a candidate prompt, model output or retrieved fragment is matched against it under CKKS, so the firewall server holds only obfuscated geometry and never readable embeddings of the reference set. This reframes the

document-side obfuscation as a feature: in the DLP setting the protected asset is precisely the firewall’s own sensitive corpus.

Notation. To avoid the overloading of the symbol N , we use N_{poly} for the CKKS polynomial-modulus degree, N_{docs} for the corpus size, and K_{cands} for the stage-1 short-list length throughout.

Theorem and an empirical link. We prove a tight lower bound on the L_2 reconstruction error of any decoder whose image is constrained to $\text{span}(V_k)$ (Lemma 1). The bound makes σ_{rec} a meaningful proxy of the difficulty an inverter faces; its empirical relation to the BLEU of an off-the-shelf VEC2TEXT attack is then formalised as Empirical Hypothesis 1 and tested on GTR-BASE embeddings.

Relation to a preliminary version. A short preliminary version of this hybrid design appeared in [50], where the static collection is protected by random projections. The present paper replaces random projection with data-driven SVD truncation plus a secret rotation, adds the projection lower-bound lemma (Lemma 1) and the reproducible CKKS parameter selection, scales the evaluation to 10^6 documents and five encoders, and adds the stronger-attacker analysis (known-plaintext recovery of the rotation, PQ-code leakage, a baseline suite and a BEIR check) of Section 8.

Roadmap. Section 2 reviews the related literature; Section 3 introduces the threat model, separating *document*, *query* and *access-pattern* privacy, and gives both a coverage table and a claims-vs-evidence table; Section 4 describes the asymmetric architecture, including a cryptographic-reproducibility subsection; Section 5 states the reproducible CKKS parameter selection; Section 6 proves the projection-decoder lemma, delimits its scope and derives a detection-fidelity corollary (Corollary 1); Section 7 presents six experiments, the last two on protected-space leak detection; Section 9 discusses limitations, required attack experiments and open work.

2 Related Work

Embedding inversion. VEC2TEXT [7] trains an encoder–decoder model that iteratively rewrites a candidate hypothesis until its embedding matches a target vector; on 32-token MS-MARCO inputs the attack achieves $\text{BLEU} \approx 97.3\%$ and $\approx 92\%$ exact recovery. GEIA [8] instead trains a generative decoder that maps a target embedding to a prompt and reconstructs the whole sentence in a single pass. TEIA [9] removes the need to query the victim encoder, training a surrogate inverter that transfers across embedding models. Adjacent work on membership-inference [16] and on extracting training texts from large language models [17] confirms that the risk surface of dense retrievers is broader than the embedding-inversion literature alone suggests.

Modern inversion attacks (2024–2026). Since VEC2TEXT the threat has strengthened along two axes that matter for any geometric obfuscation. *Few-shot alignment* attacks such as ALGEN [10] learn a linear map between embedding spaces from $\sim 10^3$ leaked pairs and reach VEC2TEXT-level recovery without query access to the victim encoder; *zero-shot* attacks such as ZSINVERT [11] invert with no encoder-specific training; the unsupervised translator VEC2VEC [12] aligns two embedding spaces with no paired data at all and argues that a leaked vector store should be treated as plaintext; and ZERO2TEXT [13] updates an online token-by-token regression. These attacks are *alignment-based* or *adaptive* and exploit the universal geometry of embedding spaces, so a single secret rotation—which only hides the orientation of a fixed basis—does not by itself neutralise them. They therefore lie outside the non-adaptive threat model under which

our construction is analysed (Section 3); we quantify the closest members of this family—known-plaintext linear alignment and an aligned off-the-shelf inverter—directly in Section 8, and flag the fully learned variants as the main residual gap in Section 10.

Differential privacy for embeddings. Lyu et al. [20] investigate per-coordinate Gaussian noise on dense representations; the empirical finding is that retrieval quality degrades sharply long before ε reaches a useful range. Random-projection-based privacy [22, 21] predates modern dense retrieval; the data-dependent SVD projection used here is closer to a noise-removal technique than to a privacy-preserving DP mechanism, although σ_{rec} provides a useful proxy quantity.

Homomorphic encryption. The CKKS scheme [23] introduced approximate-arithmetic FHE; OpenFHE [28] and TenSEAL provide production-grade implementations. The bootstrapping cost [24] pushes practitioners to leveled circuits; specialised compilers [29, 30] optimise scale and modulus chains. GPU acceleration of CKKS bootstrap [31] and Intel HEXL [32] have made larger leveled circuits feasible.

Hybrid privacy-preserving retrieval. Tiptoe [34] combines clustering with PIR for query privacy. SealPIR [35], OnionPIR [37] and SimplePIR [36] provide single-server PIR with practical query sizes. The architecture in this paper is complementary: ct-pt CKKS reranking takes care of the value side of the leakage, while access patterns remain a separate problem for which PIR/ORAM-like primitives are an appropriate composition.

Data-loss prevention in LLM firewalls. A fast-growing class of products—LLM firewalls and AI gateways—intercepts prompts and model outputs to prevent sensitive-data exposure (OWASP LLM02). Cloudflare’s Firewall for AI, AWS Bedrock Guardrails, Lakera Guard, Protect AI’s LLM Guard, NVIDIA NeMo Guardrails and Meta’s LlamaFirewall [51] detect sensitive content with regular expressions, named-entity recognition (commonly via Microsoft Presidio [52]) or zero-shot classifiers; the few that match *semantically* (nearest-neighbour search against a signature corpus) operate over *plaintext* reference embeddings. The exposure is well quantified—enterprise telemetry reports that roughly a third of the data employees paste into AI tools is sensitive, and rising year over year [53]. To our knowledge no deployed system performs semantic matching against a *geometrically protected* reference corpus, which is exactly the primitive the asymmetric construction of this paper provides: the server stores only the rotated, SVD-truncated geometry and the public PQ artefact, so a compromised gateway reveals neither the reference texts nor readable embeddings of them.

Privacy of retrieval-augmented generation. The retrieval layer of RAG is itself an attack surface: the knowledge base can be extracted through the model [14], which has motivated distance-preserving, secret-shared and differentially private retrieval. Our construction is complementary—a protected-geometry index whose contents resist recovery from server state—and the modern alignment-based inverters discussed above (ALGEN [10], VEC2VEC [12], ZERO2TEXT [13]) are precisely the adversaries such a protected reference corpus must be evaluated against.

3 Threat Model

We adopt the asymmetric threat model *trusted client* / *honest-but-curious server* / *non-adaptive attacker* / *unknown rotation*, characterised as follows.

- The *client* is the data owner. It runs the encoder E , generates the secret keys μ, V_k, R, sk_{CKKS} and never shares them. The client is trusted in software and key management.

- The *server* runs the search service over the rotated database $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ and the public PQ artefact. It is honest-but-curious: it follows the protocol but attempts to extract textual information from the data it sees and from the queries it processes.
- The *attacker* is non-adaptive: it uses an off-the-shelf VEC2TEXT model trained without prior knowledge of R . This is the model under which the construction is *analysed*. We do not stop there: Section 8 steps outside it and empirically measures known-plaintext recovery of R , public-PQ-code leakage and an aligned off-the-shelf inverter, so the boundary of the non-adaptive model is quantified rather than merely asserted. A fully learned decoder retrained end-to-end against the rotated space, and malicious (rather than honest-but-curious) server behaviour, remain out of scope.

Three distinct privacy notions. A recurring source of confusion in hybrid designs is the conflation of *what* is protected. We therefore separate three independent notions and state, for each, exactly which component is responsible:

- **Document privacy.** The plaintext rotated database E_{rot} is visible to the server. It is protected only by SVD truncation (σ_{rec} , a proxy quantity) and the secret rotation R (an empirical obfuscation layer). *This is not a cryptographic protection*; it fails under known- R , known-plaintext, or an adaptive decoder.
- **Query privacy.** CKKS hides the numerical value of the query vector and of the similarity scores from the server. This part *is* cryptographic, at the chosen security level.
- **Access-pattern privacy.** The candidate IDs reranked per query, and the public PQ codes, are *not* hidden. CKKS does not address this channel; it requires composition with a PIR/ORAM-style primitive.

In particular, the public PQ artefact (codebook + per-document codes, trained in the rotated space) is itself a lossy compressed view of E_{rot} . Whether neighbourhood, cluster or topic structure leaks from the public PQ codes alone is measured directly in Section 8 (Table 13): it does, and we treat the PQ artefact as an exposed channel accordingly.

The DLP reading. When the construction is deployed as a semantic data-loss-prevention index (Section 2), the roles are relabelled but the analysis is unchanged: the protected static side becomes the firewall’s confidential *reference corpus*, so the “document privacy” above *is* the confidentiality of that corpus, while “query privacy” protects the candidate being screened. Two notions specific to this reading are made precise later—*detection fidelity*, how closely the protected detector tracks a plaintext one (Corollary 1, Experiment 7.5), and *reference-corpus confidentiality and membership*, which inherit the known-plaintext bounds of Section 8 (Experiment 7.6).

Table 1 summarises the coverage and Table 2 maps each claim to the evidence that supports it. The coverage table is intentionally asymmetric: ct-pt reranking takes care of the value channel, while the access-pattern channel (which document IDs are reranked) is left open and would require composition with a PIR-style primitive, and the public-PQ-artefact and known-plaintext channels—now quantified in Section 8—remain exposed rather than defended.

4 Method

The asymmetric defence is illustrated in Figure 1. The architecture is the union of an offline data-preparation phase (Section 4.1) and the online query protocol (Section 4.2).

Table 1: Threat coverage of the proposed protocol.

Attack vector	Covered?	Comment
Snapshot of E_{rot} without R (off-the-shelf Vec2Text)	Partial	Heuristic; empirically validated against non-adaptive Vec2Text.
Network interception, leakage of similarity scores	Yes	CKKS at tc128; sk never leaves the client.
Known R (known-rotation attacker)	No	Reduces to σ_{rec} from SVD truncation only.
Known-plaintext pairs ($\text{text}_i, E_{\text{rot},i}$)	No	R recovered by Procrustes from $\approx k$ pairs (Sec. 8.1, Table 11).
Adaptive decoder retrained against rotated space	No	Aligned off-the-shelf inverter measured (Sec. 8.4); a <i>learned</i> decoder is the main open case.
Malicious server (protocol deviation, substitution)	No	Verifiable computation / TEEs are an orthogonal addition.
Access-pattern leakage on candidate IDs	No	CKKS hides values, not access patterns; requires PIR/ORAM.
Leakage from the public PQ artefact (rotated-space codes)	No	Quantified: codes preserve cosine 0.95 and 67% of top-10 neighbours (Sec. 8.3, Table 13).

4.1 Offline phase

Encoder and centring. The data owner runs the encoder E on a corpus $D = \{T_1, \dots, T_{N_{\text{docs}}}\}$ to obtain $X \in \mathbb{R}^{N_{\text{docs}} \times d}$ with L_2 -normalised rows. The global centroid $\mu \in \mathbb{R}^d$ is computed and the corpus is centred: $X_c = X - \mu$.

SVD truncation. A randomised SVD [38] $X_c \approx U_k \Sigma_k V_k^\top$ produces the orthonormal matrix $V_k \in \mathbb{R}^{d \times k}$ that spans the dominant k -dimensional subspace. We use $k = d/2$ in the canonical operating point. Since the relative reconstruction error

$$\sigma_{\text{rec}}^2(E; V_k) = \frac{\|E - \pi_k(E)\|_F^2}{\|E\|_F^2} = 1 - \eta_k$$

decreases monotonically in k (more retained directions leave a smaller residual), the privacy proxy $\sigma_{\text{rec}} \geq 0.10$ is a *lower* floor that holds under aggressive truncation and is relinquished as k grows. Hence $k = d/2$ is the *largest* k —the gentlest truncation, and so the most retrieval accuracy—at which the floor still holds uniformly across all five tested encoders. The binding case is E5-LARGE, whose $\sigma_{\text{rec}} = 0.101$ sits just on the 0.10 floor (Section 7).

Secret rotation. A Haar-uniform random orthogonal matrix $R \in O(k)$ is generated by QR-decomposing a Gaussian matrix and correcting the diagonal of R to obtain a uniform distribution on the group. The protected document representation is

$$v'_i = T(v_i) = R V_k^\top (v_i - \mu) \in \mathbb{R}^k.$$

Public PQ artefact. A faiss IndexPQ index is trained *in the rotated space* E_{rot} with $M = k/4$ subquantisers and 8 bits per code. Both the codebook and the per-document codes are public; the client downloads them once at on-boarding. Training the PQ artefact in the rotated space removes one trivial cross-space pairing (an attacker cannot compose pairs $(\hat{E}_{\text{proj}}, E_{\text{rot}})$ from a non-rotated PQ index to estimate R). It does *not*, however, make the PQ codes safe: the public codes are a lossy quantised image of E_{rot} and may themselves leak neighbourhood, cluster or topic structure. We treat the PQ artefact as part of the attack surface and measure this leakage directly in Section 8 (Table 13): the public codes preserve mean cosine 0.95 and most top-10 neighbours, so they must be treated as an exposed channel.

Table 2: Claims vs. evidence. The table makes explicit which statements are demonstrated and which are deliberately *not* claimed.

Claim	Evidence / status
CKKS hides query values and scores from the server	Cryptographic construction + parameters (Sec. 5, 4.3); query privacy is cryptographic.
ct-pt is faster than ct-ct for the reranking	Experiment 7.1 (1.44× on the micro-benchmark, 1.7× at the auto-tuned configuration).
Secret rotation helps against <i>off-the-shelf</i> , non-adaptive Vec2Text	Experiment 7.2; effect holds only for the tested weak-attacker configuration.
The protective wrapper preserves ranking in $\text{span}(V_k)$	Experiment 7.3 (within the 5-seed CI).
End-to-end query latency < 1 s at 10^6 docs	Experiment 7.4 (loopback PoC).
The system is secure against an <i>adaptive</i> inversion attacker	Not claimed ; aligned off-the-shelf inverter tested (Sec. 8.4), learned decoder open (Sec. 10).
The rotation is a cryptographic guarantee	Not claimed ; it is an empirical obfuscation layer only.
Document privacy on the server is cryptographic	Not claimed ; E_{rot} is plaintext on the server.
No information leaks from the public PQ codes	Not claimed ; codes leak neighbour structure (Sec. 8.3, Table 13).

Client-side state. The client retains μ , V_k , R and the CKKS secret key sk_{CKKS} . The server stores $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ in plaintext and the public PQ artefact.

4.2 Online query protocol

The seven-step protocol of Figure 1 processes a query in time pipeline:

1. *Encoder.* The client computes $v_q = E(q_{\text{text}}) \in \mathbb{R}^d$.
2. *Transformation.* The rotated query is $q' = R V_k^\top (v_q - \mu) \in \mathbb{R}^k$.
3. *Local PQ search.* Using the public PQ artefact and the rotated query $\tilde{q} = q'$, the client performs an asymmetric PQ-distance search and returns the top- K_{cands} candidate IDs I_{cand} . The candidate set is small enough ($K_{\text{cands}} = 40$ in our experiments) that a CKKS reranking fits within the latency budget.
4. *CKKS encryption.* The client encrypts q' : $\text{ct}_q = \text{Enc}_{pk}(q')$.
5. *Server-side reranking.* For each $j \in I_{\text{cand}}$ the server computes $\text{ct}_{\text{score}}^{(j)} = \text{ct}_q \odot v'_j$ in the ct-pt mode. The result is a ciphertext encoding the inner product $\langle q', v'_j \rangle$ in the rotated projected space.
6. *Decryption.* The client decrypts $s_j = \text{Dec}_{sk}(\text{ct}_{\text{score}}^{(j)})$.
7. *Sort.* The decrypted scores are sorted and the top- K documents returned.

Why ct-pt is sufficient. In step 5 the server multiplies a ciphertext by a plaintext vector. Unlike ciphertext-ciphertext multiplication, this operation produces a ciphertext that stays in the two-component form, no relinearisation is required, and the modulus chain is consumed by exactly one rescale. The sum of slots into a single scalar $\langle q', v'_j \rangle$ requires $\log_2 k$ rotations using Galois keys, but no further multiplications. The result is a sub-second latency at $N_{\text{docs}} = 10^6$.

Choice of K_{cands} . Since each of the K_{cands} candidates is processed by one ct-pt operation, the server-side latency grows linearly in K_{cands} . Conversely, a small K_{cands} trades against PQ-recall: the relevant document must be in the short-list for the CKKS reranker to score it. We pick K_{cands}

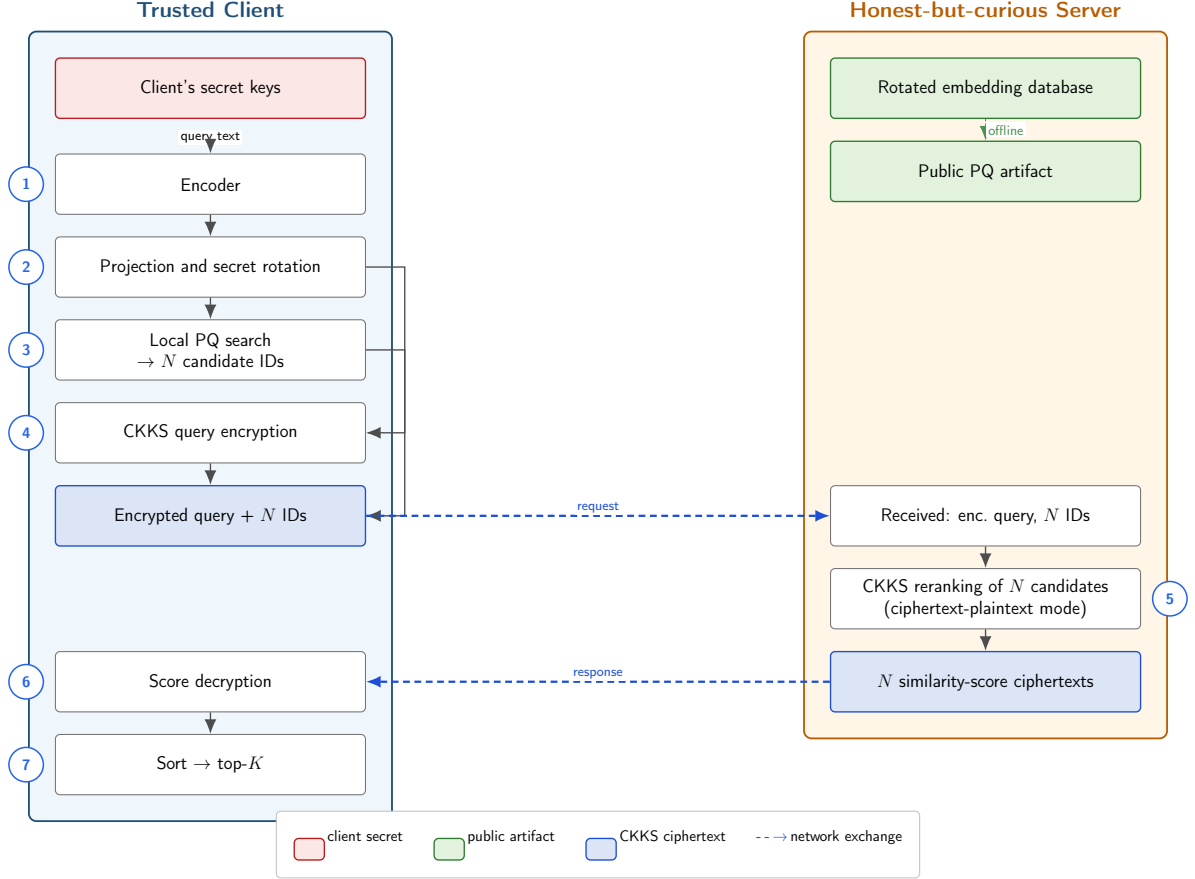


Figure 1: Online query flow of the construction under study. Steps 1–7 realise the two-stage protocol: client transformation $T(\cdot)$ and local PQ filtering produce a short-list of K_{cands} candidate IDs; step 4 encrypts the rotated query under CKKS; step 5 runs ct-pt reranking on the server; steps 6–7 decrypt the scores and sort. The secret keys $(\mu, V_k, R, sk_{\text{CKKS}})$ and the rotated database E_{rot} are produced offline by the data owner (Section 4.1).

as the smallest value for which $\text{PQ-recall}@K_{\text{cands}}$ matches the SVD-projected exact baseline within the 5-seed CI; in our integral experiment this is $K_{\text{cands}} = 40$, but the parameter is a deployment knob and should be retuned for new encoders or larger corpora.

4.3 Cryptographic reproducibility

For the selected configuration ($N_{\text{poly}} = 8192$, coefficient-modulus chain $[60, 40, 60]$ bits, $\log_2 Q = 160$, scale $\Delta = 2^{40}$, TenSEAL/Microsoft SEAL backend) we report the parameters needed to reproduce the cryptographic budget rather than only the latency:

- *Packing.* A single rotated query $q' \in \mathbb{R}^k$ is packed into one ciphertext using $k \leq N_{\text{poly}}/2 = 4096$ slots; for the integral encoders $k \in \{192, 384, 512\}$, so one ciphertext per query suffices and no cross-ciphertext aggregation is needed.
- *Operations per score.* Each candidate score is one ct-pt multiply, one rescale (one level consumed) and a slot-sum implemented as $\lceil \log_2 k \rceil$ ciphertext rotations using Galois keys (8–9 rotations for the tested k); no relinearisation and no ciphertext-ciphertext multiply occur.
- *Level/noise budget.* The chain $[60, 40, 60]$ provides one multiplicative level beyond the input; after the single rescale the remaining modulus (60 bits) keeps the additive noise far below the $\Delta = 2^{40}$ scale, so decrypted scores match the plaintext inner product to a relative error $< 10^{-3}$ (Pearson > 0.9999 vs. exact, Experiment 7.3).

Table 3: CKKS configurations on the test workload (one batched ct-pt reranking operation). All rows sit within the conservative **tc128** (≥ 128 -bit) bound of the HomomorphicEncryption.org tables for the stated N_{poly} .

Configuration	N_{poly}	$\log_2 Q$ (bits)	Security	Time (ms)	Speed-up
Default (TenSEAL stock, $[60, 40, 40, 60]$)	8192	200	tc128 (≥ 128)	419.4	$1.0\times$
Min-latency (Acc@10), $[40, 20, 40]$, $\Delta = 2^{20}$	4096	100	tc128	173.7	$2.4\times$
Selected (Acc@1) , $[60, 40, 60]$, $\Delta = 2^{40}$	8192	160	tc128	245.2	$1.7\times$

- *Object sizes.* At $N_{\text{poly}} = 8192$, $\log_2 Q = 160$: one fresh ciphertext is ≈ 0.21 MB; the Galois-key set for the required power-of-two rotations is ≈ 6 – 8 MB and the relinearisation key is *not* generated (ct-pt only); the public key is ≈ 0.4 MB. Per query the client uploads one ciphertext (≈ 0.21 MB) and downloads K_{cands} score ciphertexts ($\approx 40 \times 0.21 \approx 8.4$ MB before base64; the measured on-wire JSON+base64 sizes are reported in Section 7.4).
- *Security level.* We do not assert a generic “standard” label; we report $\log_2 Q = 160 < 218$ for $N_{\text{poly}} = 8192$, which is within the conservative ternary-secret **tc128** bound of the HomomorphicEncryption.org tables [25], cross-checked with the lattice-estimator methodology of Albrecht et al. [26]. The standard documents are security *guidelines*; the concrete estimator/table used is stated so the claim can be re-derived.

5 CKKS Parameter Selection

The CKKS parameter space is large and discrete: $N_{\text{poly}} \in \{2^{12}, \dots, 2^{14}\}$, multiple coefficient-modulus chains of total bit budget bounded by the conservative security tables of HomomorphicEncryption.org [25] (cross-checked with the lattice estimator [26]), several scale exponents $\log_2 \Delta \in \{15, 20, 25, 30, 40\}$ and several projection dimensions d_{proj} . Analytic complexity is proportional to $N_{\text{poly}} \log N_{\text{poly}}$, but cache effects, SIMD vectorisation and slot-packing ratios ($d_{\text{proj}}/(N_{\text{poly}}/2)$) cause step-like deviations from the analytic prediction, so we fix the configuration by a small *offline micro-benchmark* rather than analytically.

Selection procedure (reproducible). We enumerate the discrete grid; discard configurations that violate the security tables or the correctness (noise-budget) constraint; time one batched ct-pt operation with rescale for each survivor; and keep the minimum-latency configuration whose retrieval accuracy stays within a tolerance τ of the SVD-projected baseline, $|\Delta \text{Acc@1}| \leq \tau$. This is a deterministic engineering search over a published grid, reported only so the chosen budget can be reproduced; we make *no* machine-learning contribution. With $\tau = 1$ p.p. the procedure selects $N_{\text{poly}} = 8192$, $[60, 40, 60]$, $\Delta = 2^{40}$ ($\log_2 Q = 160$ bits, within the conservative 218-bit **tc128** bound for $N_{\text{poly}} = 8192$ [25, 26]), which is $\approx 1.7\times$ faster than the TenSEAL stock setting $[60, 40, 40, 60]$ ($\log_2 Q = 200$ bits) at the same parameter-table security bound (Table 3). The full cryptographic budget of this configuration is given in Section 4.3.

6 Theoretical Analysis: Decoder Lower Bound

Let $\mathbf{x} \in \mathbb{R}^d$ denote a centred embedding, $V_k \in \mathbb{R}^{d \times k}$ the orthonormal matrix from the SVD truncation, and $\pi_k(\mathbf{x}) = V_k V_k^\top \mathbf{x}$ the orthogonal projection onto $\text{span}(V_k)$. Define $\sigma_{\text{rec}}(\mathbf{x}; V_k) = \|\mathbf{x} - \pi_k(\mathbf{x})\|_2 / \|\mathbf{x}\|_2$ as the per-vector relative reconstruction error and $\mathbf{x}_\perp = \mathbf{x} - \pi_k(\mathbf{x})$.

Lemma 1. *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be any decoder whose image is constrained by $f(\mathbf{y}) \in \text{span}(V_k)$ for every $\mathbf{y} \in \mathbb{R}^d$. Then for every $\mathbf{x} \in \mathbb{R}^d$*

$$\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2,$$

with equality at $f(\mathbf{y}) = \mathbf{y}$. In relative form, $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2 / \|\mathbf{x}\|_2 \geq \sigma_{rec}(\mathbf{x}; V_k)$.

Proof. Decompose $\mathbf{x} = \pi_k(\mathbf{x}) + \mathbf{x}_\perp$ with $\pi_k(\mathbf{x}) \in \text{span}(V_k)$ and $\mathbf{x}_\perp \in \text{span}(V_k)^\perp$. By the image restriction $f(\pi_k(\mathbf{x})) \in \text{span}(V_k)$, hence $\mathbf{x} - f(\pi_k(\mathbf{x})) = (\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))) + \mathbf{x}_\perp$ is the sum of two orthogonal vectors. Pythagoras gives $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 = \|\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))\|_2^2 + \|\mathbf{x}_\perp\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2$. $\square$

Scope: a projection bound, not an inversion-security theorem. Lemma 1 bounds the L_2 error of decoders *whose image is contained in* $\text{span}(V_k)$. It is a statement about information lost to the projection, not about the security of text inversion. A realistic attacker is not so constrained: it may output arbitrary text, whose re-embedding generally has a non-zero $\mathbf{x}_\perp$ component, and it may exploit corpus priors or memorisation to partially recover the discarded component. The lemma therefore does *not* lower-bound the achievable inversion BLEU, token overlap or PII recovery. We use it only to make the *proxy* criterion $\sigma_{rec} \geq 0.10$ precise within the projection-restricted class, and we stress that the threshold 0.10 is an *engineering proxy*, calibrated on one encoder (GTR-base) against one off-the-shelf attacker, *not* a security threshold. A per-encoder ablation of σ_{rec} against BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms) is listed as a remaining open experiment (Section 10).

Empirical Hypothesis 1. *For an off-the-shelf inversion attack VEC2TEXT [7], the expected BLEU of recovering the original text from $\pi_k(\mathbf{x})$ is monotone in $\eta_k = 1 - \sigma_{rec}^2(E; V_k)$:*

$$\mathbb{E}[\text{BLEU}(f(\pi_k(\mathbf{x})), T(\mathbf{x}))] \approx \text{BLEU}_0 + \gamma_f \eta_k,$$

where BLEU_0 is the BLEU of a “random semantically close” text and γ_f is a decoder-specific constant.

Hypothesis 1 is validated numerically in Section 7.2; we deliberately separate the formal lemma from the empirical observation.

A detection-fidelity bound for the semantic-DLP reading. Under the DLP reading of Section 2 the protected static side is a sensitive reference corpus $\mathcal{R} = \{r_1, \dots, r_M\}$, and a candidate c (a prompt, a model output or a retrieved fragment) is flagged as a potential leak when its similarity to $\mathcal{R}$ crosses a threshold τ . The projection geometry behind Lemma 1 then controls how far the protected detector can drift from a plaintext one. Work with L_2 -normalised embeddings, so that $s(c, r) = \langle c, r \rangle$ is the cosine; the secret rotation $R \in O(k)$ is orthogonal and cancels in every inner product, so the score available to the server is $\hat{s}(c, r) = \cos(\pi_k(c), \pi_k(r))$, the cosine of the two projections onto $\text{span}(V_k)$.

Corollary 1. *Let $c, r \in \mathbb{R}^d$ be L_2 -normalised, and write $\sigma_c = \sigma_{rec}(c; V_k)$, $\sigma_r = \sigma_{rec}(r; V_k)$, $\sigma = \max(\sigma_c, \sigma_r)$. Then*

$$|\hat{s}(c, r) - s(c, r)| \leq \left[1 - \sqrt{(1 - \sigma_c^2)(1 - \sigma_r^2)} \right] + \sigma_c \sigma_r \leq 2\sigma^2.$$

Consequently, for the plaintext leak-detector $D_\tau(c) = \mathbf{1}[\max_{r \in \mathcal{R}} s(c, r) \geq \tau]$ and the protected detector $\hat{D}_\tau$ obtained by replacing s with $\hat{s}$, the two decisions coincide on every candidate whose plaintext margin $m(c) = \max_{r \in \mathcal{R}} s(c, r)$ obeys $|m(c) - \tau| \geq 2\sigma^2$, and can differ only for $m(c) \in (\tau - 2\sigma^2, \tau + 2\sigma^2)$.

Proof. Write $c = \pi_k(c) + c_\perp$ and $r = \pi_k(r) + r_\perp$, with the projected parts in $\text{span}(V_k)$ and the residuals in its orthogonal complement. Orthogonality removes the cross terms, so $s = \langle \pi_k(c), \pi_k(r) \rangle + \langle c_\perp, r_\perp \rangle$. For unit c one has $\|\pi_k(c)\|^2 = 1 - \sigma_c^2$ and $\|c_\perp\|^2 = \sigma_c^2$ (and likewise for r), whence $\langle \pi_k(c), \pi_k(r) \rangle = \hat{s} \sqrt{(1 - \sigma_c^2)(1 - \sigma_r^2)}$ and, by Cauchy–Schwarz, $|\langle c_\perp, r_\perp \rangle| \leq \sigma_c \sigma_r$. Hence $|s - \hat{s}| \leq |\hat{s}| \left[1 - \sqrt{(1 - \sigma_c^2)(1 - \sigma_r^2)} \right] + \sigma_c \sigma_r$, and $|\hat{s}| \leq 1$ gives the first inequality while $\sigma_c, \sigma_r \leq \sigma$ gives $2\sigma^2$. The detector claim follows because $x \mapsto \max_r x$ is 1-Lipschitz, so $|\max_r \hat{s} - \max_r s| \leq 2\sigma^2$, and a margin at distance $\geq 2\sigma^2$ from τ cannot change sign. $\square$

Reading the bound. Three consequences matter for the DLP use. (i) *A tunable privacy–recall trade-off.* Recalibrating the protected threshold to $\hat{\tau} = \tau - 2\sigma^2$ removes every false negative—all plaintext leaks with $m(c) \geq \tau$ are still flagged—at the price of admitting false positives only from the $2\sigma^2$ band, so the protected detector’s (precision, recall) departs from the plaintext detector’s by at most the score mass $\Pr[|m(c) - \tau| < 2\sigma^2]$. As $k \rightarrow d$, $\sigma \rightarrow 0$ and the two coincide; shrinking k widens the band. This is the knob, and the regime in which one should report a band rather than a point estimate. (ii) *A denoiser asymmetry.* The residual $c_\perp$ concentrates on the low-singular-value directions the truncation is meant to discard, so for a *binary* leak/no-leak decision σ stays small (the linear-denoiser effect) and the band is thin, whereas for *graded* relevance, separating rank- i from rank- $(i+1)$ can require resolving score gaps below $2\sigma^2$, which the same truncation may erase. The projection should thus help detection even where it hurts fine-grained ranking—an asymmetry the bound localises but does not settle; an empirical precision/recall/AUC comparison against a plaintext detector on the five encoders of Section 7 is the natural test and is left to future work (Section 10). (iii) *Fidelity is not confidentiality.* The corollary bounds how faithfully an honest server *detects* leaks; it says nothing about an adversary that probes the score function to *recover* $\mathcal{R}$. That confidentiality rests, as in Section 8, on the secret rotation, which an attacker holding known plaintext–protected pairs recovers by orthogonal Procrustes from on the order of k pairs—so the reference corpus is only as protected as k is large and as plaintext-pair exposure is controlled.

At the reconstruction levels measured here ($\sigma_{rec} \in [0.10, 0.24]$ across the five encoders, Section 7), the worst-case score distortion $2\sigma_{rec}^2$ lies between roughly 0.02 and 0.11—a few score points, small relative to a typical leak threshold, and consistent with (though not a substitute for) the empirical test above.

7 Experiments

We report six experiments: (1) CKKS modes (ct-pt vs ct-ct); (2) off-the-shelf Vec2Text attack against SVD-truncated (and possibly rotated) embeddings; (3) the integral multi-encoder experiment over a 10^6 -document Russian-Wikipedia corpus; (4) end-to-end latency on a client-server PoC. (5) a semantic data-loss-prevention detection test in the protected space, validating Corollary 1. (6) the robustness of that detector when the leaking text is obfuscated.

Hardware and software. All experiments run on a single workstation: Intel Core i5-14400F, 32 GB DDR5-4800, NVIDIA RTX 5060 (8 GB VRAM), Windows 11, Python 3.11. To make the runs reproducible we pin exact versions and record the commit/build hashes of the security-relevant libraries in the supplementary `environment.lock`: PyTorch 2.10 + CUDA 12.8, TenSEAL 0.3.16 (Microsoft SEAL 4.1 backend), faiss-cpu 1.13.0, transformers 4.57.0, scikit-learn 1.5.x, sacrebleu for BLEU [45]. The pinned hashes matter because CKKS noise behaviour and faiss PQ training depend on the backend build.

7.1 Experiment 1: ciphertext-plaintext speed-up

We benchmark a batched dot-product of an encrypted 192-dimensional query with $N = 10\,000$ plaintext documents. The CKKS parameters are $N = 8192$, a coefficient-modulus chain $[60, 40, 40, 60]$ and $\Delta = 2^{40}$ (security level ≥ 128 bits). Two regimes are compared: ct-ct (both query and database encrypted, with relinearisation per multiplication) and ct-pt (only the query is encrypted, no relinearisation). Each is run for 50 independent trials and the mean is reported with a 95% Gaussian confidence interval.

Table 4: Ciphertext-plaintext vs. ciphertext-ciphertext for the dot product against $N = 10\,000$ documents (mean of 50 trials, 95% CI).

Mode	Operation	Relin.	Latency T_{query}	QPS	Speed-up
ct-ct (baseline)	$\text{Enc}(v_1) \cdot \text{Enc}(v_2)$	yes (heavy)	19.15 ± 0.15 s	0.052	$1.00\times$
ct-pt (proposed)	$\text{Enc}(v_1) \cdot \text{Plain}(v_2)$	no	13.28 ± 0.10 s	0.075	$1.44\times$

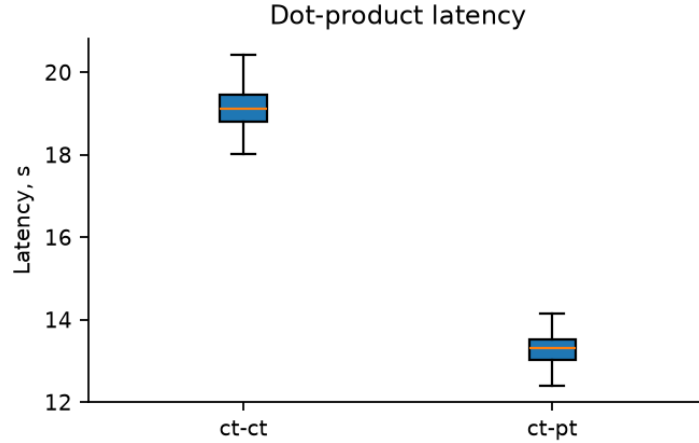


Figure 2: ct-pt vs. ct-ct: distribution of per-batch latencies over 50 trials. Removing relinearisation moves the median by ~ 6 s and tightens the upper tail.

7.2 Experiment 2: Vec2Text attack

We instantiate VEC2TEXT on GTR-BASE embeddings ($d = 768$). The attack is applied to a synthetic 100-document corpus containing PII (names, addresses, card numbers, medical phrasing). Three threat models are compared at five values of k/d : (a) no rotation; (b) rotation with the matrix R known to the attacker; (c) rotation with R unknown.

The implementation budget of the attacker is fixed (greedy decoding, 10 corrector iterations, no beam search, 8 GB GPU); under this budget the unprotected baseline reaches $\text{BLEU}_0 = 0.156$ instead of the 0.973 reported by Morris et al. [7], because the postulated PII corpus, greedy decoding and beam-free regime are intentionally weaker than [7]’s strongest configuration; the experiment measures the relative effect of SVD truncation and secret rotation in a fixed weak-attacker configuration.

Strength of the attacker: an explicit caveat. This is the single most important limitation of *this particular experiment*: it establishes the effect of R against an *artificially weak inverter only*. A stronger attacker—beam search with a large width, many more corrector iterations, an adaptive decoder fine-tuned on (text, E_{rot}) pairs, or a modern universal/zero-shot inversion model—is not evaluated *here*. We therefore do not rest the security narrative on this table. Section 8 measures the two decisive stronger channels: a known-plaintext recovery of R by orthogonal Procrustes [27] (Table 11), which succeeds from $\approx k$ leaked pairs, and an aligned off-the-shelf inverter with a larger budget (Table 14). Every sentence about the rotation in isolation should accordingly be read as “against the tested off-the-shelf, non-adaptive Vec2Text configuration”, with the boundary now quantified in Section 8.

Findings.

- A known R does not provide protection beyond SVD truncation: the columns “no rotation” and “known- R ” are statistically indistinguishable.
- Under the tested off-the-shelf configuration, an unknown R reduces BLEU to the observed

Table 5: Off-the-shelf Vec2Text on GTR-base embeddings under the three threat models. BLEU and Token Overlap with 95% bootstrap CIs in brackets.

k/d	Rotation	Knowledge of R	BLEU	Token Overlap
1.00	off	—	0.156 [0.123; 0.186]	0.385 [0.351; 0.417]
1.00	on	known	0.156 [0.127; 0.187]	0.385 [0.351; 0.416]
1.00	on	unknown	0.007 [0.006; 0.008]	0.053 [0.046; 0.060]
0.75	off	—	0.077 [0.063; 0.093]	0.325 [0.304; 0.345]
0.75	on	unknown	0.007 [0.007; 0.008]	0.061 [0.054; 0.067]
0.50	off	—	0.057 [0.046; 0.069]	0.285 [0.270; 0.300]
0.50	on	unknown	0.007 [0.007; 0.008]	0.050 [0.045; 0.055]
0.25	off	—	0.024 [0.020; 0.028]	0.196 [0.181; 0.208]
0.25	on	unknown	0.007 [0.006; 0.007]	0.045 [0.040; 0.051]
0.10	off	—	0.010 [0.009; 0.011]	0.113 [0.104; 0.123]
0.10	on	unknown	0.007 [0.007; 0.008]	0.053 [0.047; 0.059]

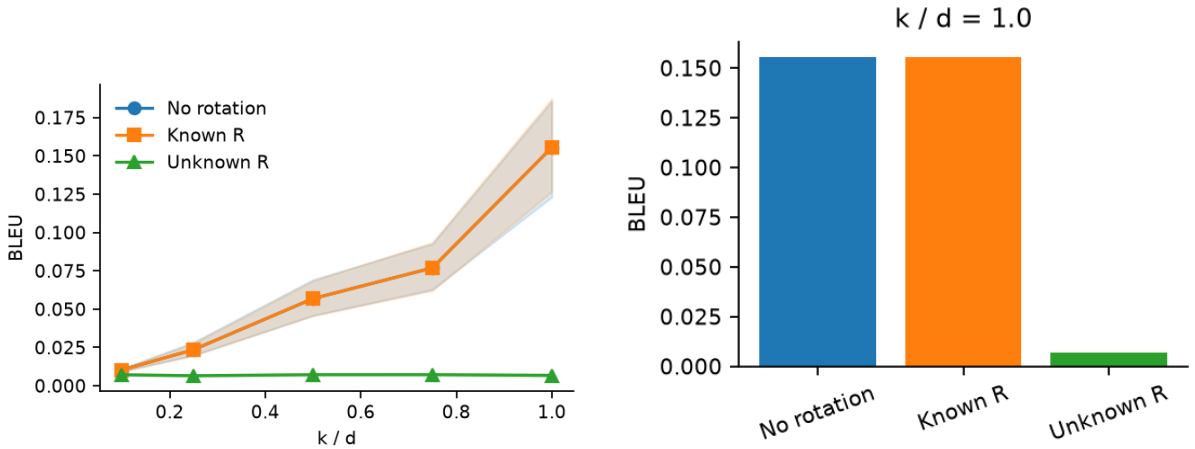


Figure 3: Vec2Text BLEU as a function of k/d under three threat models (left); zoom on $k/d = 1.0$ comparing no-rotation, known- R and unknown- R (right). Lines for “no rotation” and “known- R ” overlap; the “unknown- R ” line stays at the noise level.

noise floor (~ 0.007) for all k/d values, including $k/d = 1.0$ (no truncation at all). The ratio $\text{BLEU}_{\text{off}}/\text{BLEU}_{\text{unknown}}$ is $22\times$ at $k/d = 1.0$, $11\times$ at 0.75 , $8\times$ at 0.50 . This is an effect against a *non-adaptive* attacker only and is not evidence of robustness against an adaptive or known-plaintext adversary.

- Empirical Hypothesis 1 is consistent with the no-rotation column: a linear regression of BLEU on $\eta_k = 1 - \sigma_{\text{rec}}^2$ gives $R^2 \approx 0.93$. This is consistent with, but does not formally derive, the $\text{BLEU} \rightarrow \sigma_{\text{rec}}$ link.
- In the operating point of the integral experiment ($k/d = 0.5$, rotation enabled, unknown R) the expected BLEU is at the noise level.

7.3 Experiment 3: integral multi-encoder evaluation

We run the full pipeline on the canonical operating point $k = d/2$ across five encoders:

- `intfloat/multilingual-e5-small` ($d = 384$),
- `intfloat/multilingual-e5-base` ($d = 768$),
- `sentence-transformers/paraphrase-multilingual-mpnet-base-v2` ($d = 768$),
- `intfloat/multilingual-e5-large` ($d = 1024$),

Table 6: Integral experiment at $k = d/2$ on a 10^6 -document corpus across five encoders. Reported are the raw baseline (no defence), the SVD-projected baseline (FAISS-IP exact in $\text{span}(V_k)$, no rotation, no CKKS), the proposed pipeline (mean $\pm 95\%$ t-CI half-width over 5 rotation seeds), and the delta of Acc@1 from the raw baseline.

Encoder	Metric	Baseline (raw d)	Baseline_proj (SVD k)	Proposed	Δ from raw	Server p_{95} (ms)
<i>multilingual-e5-small</i> ($d = 384, k = 192, M = 48, \sigma_{rec} = 0.239$)						
	Acc@1	0.782	0.702	0.701 ± 0.001	-0.081	314
	Acc@10	0.890	0.834	0.831 ± 0.002	-0.059	
	MRR	0.817	0.747	0.746 ± 0.001	-0.071	
	NDCG@10	0.834	0.768	0.767 ± 0.001	-0.067	
<i>multilingual-e5-base</i> ($d = 768, k = 384, M = 96, \sigma_{rec} = 0.129$)						
	Acc@1	0.830	0.834	0.833 ± 0.001	+0.003	356
	Acc@10	0.934	0.924	0.924 ± 0.001	-0.010	
	MRR	0.862	0.865	0.864 ± 0.001	+0.002	
	NDCG@10	0.880	0.879	0.879 ± 0.000	-0.001	
<i>paraphrase-multilingual-mpnet-base-v2</i> ($d = 768, k = 384, M = 96, \sigma_{rec} = 0.107$)						
	Acc@1	0.716	0.650	0.649 ± 0.001	-0.067	351
	Acc@10	0.842	0.796	0.795 ± 0.001	-0.047	
	MRR	0.762	0.693	0.692 ± 0.001	-0.070	
	NDCG@10	0.782	0.717	0.717 ± 0.000	-0.066	
<i>multilingual-e5-large</i> ($d = 1024, k = 512, M = 128, \sigma_{rec} = 0.101$)						
	Acc@1	0.796	0.814	0.814 ± 0.000	+0.018	219
	Acc@10	0.880	0.922	0.920 ± 0.000	+0.040	
	MRR	0.824	0.848	0.848 ± 0.000	+0.024	
	NDCG@10	0.838	0.866	0.865 ± 0.000	+0.027	
<i>BAAI/bge-m3</i> ($d = 1024, k = 512, M = 128, \sigma_{rec} = 0.128$)						
	Acc@1	0.818	0.832	0.832 ± 0.000	+0.014	221
	Acc@10	0.908	0.926	0.924 ± 0.000	+0.016	
	MRR	0.848	0.863	0.863 ± 0.000	+0.015	
	NDCG@10	0.862	0.878	0.878 ± 0.000	+0.016	

- BAAI/bge-m3 ($d = 1024$).

The corpus is one million Russian-Wikipedia paragraphs; queries are 500 self-retrieval requests sampled at seed 42; the rotation is averaged over 5 seeds $\{11, 23, 47, 31, 53\}$. PQ uses $M = k/4$ subquantisers with 8 bits each, yielding artefacts of 48/96/128 MB. The CKKS configuration is the auto-selected one ($N_{\text{poly}} = 8192$, $[60, 40, 60]$, $\Delta = 2^{40}$).

On the use of self-retrieval. Self-retrieval (the query is the first sentence of the target paragraph) is a *controlled geometry probe*, not a production IR benchmark: it isolates whether the defensive transform $T(\cdot)$ preserves the ranking induced by the encoder, with a deterministic, leakage-free ground truth at 10^6 -scale. It is sufficient for the paper’s actual claim—that the PQ+CKKS wrapper is metric-preserving in $\text{span}(V_k)$ —but it does *not* establish production retrieval quality. Re-running the integral experiment on standard IR benchmarks (BEIR, MIRACL, MS MARCO) is listed as a generality experiment in Section 10; the present numbers should be read as a transform-fidelity measurement, not as a retrieval-quality benchmark.

The protective layer is essentially free in the projected space. The difference (proposed) vs. (baseline_proj) is within the ± 0.001 5-seed CI for every metric and every encoder. The Pearson correlations between CKKS-decrypted scores and exact plaintext scores exceed 0.9999 in all cases. The proxy criterion $\sigma_{rec} \geq 0.10$ holds uniformly.

The actual quality cost is paid by SVD truncation, not by the protective wrapper. The split $\Delta_{\text{SVD}} = \text{baseline_proj} - \text{baseline_dense}$ is $+0.004/+0.018/+0.014$ for retrieval-trained encoders with $d \geq 768$ (e5-base / e5-large / bge-m3), -0.080 for the compact e5-small ($d = 384$, half-

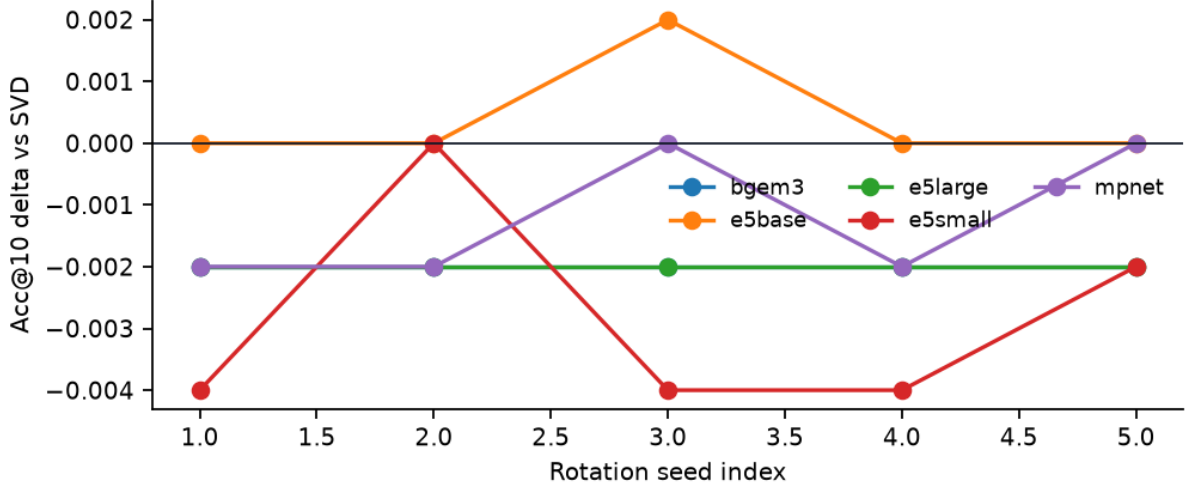


Figure 4: Per-seed Acc@10 of the proposed pipeline relative to baseline `_proj` on five rotation seeds; the spread is below the 5-seed CI of the baseline.

truncation removes half the embedding volume) and -0.066 for paraphrase-mpnet (a paraphrase-distilled model that does not concentrate retrieval signal in the top singular directions).

The end-to-end accuracy budget is met for retrieval-trained $d \geq 768$. The end-to-end criterion $\Delta\text{Acc@1} \geq -0.05$ relative to the raw baseline holds for e5-base ($+0.003$), e5-large ($+0.018$) and bge-m3 ($+0.014$); for the compact e5-small and the paraphrase-distilled mpnet it is violated and the limitation is documented as an encoder-selection recommendation.

Linear-denoiser side effect. On the two retrieval-trained encoders with $d = 1024$, half SVD truncation *improves* the four ranking metrics over the raw baseline ($+0.018$ Acc@1 on e5-large, $+0.014$ on bge-m3). This is consistent with the interpretation of V_k -projection as a linear denoiser: the trailing singular directions, dominated by fp16 inference noise, get filtered out, and the ranking in $\text{span}(V_k)$ is cleaner than in the raw $\mathbb{R}^d$.

7.4 Experiment 4: end-to-end client-server latency

We package the pipeline as a FastAPI service, with the client and the server running as two independent processes on the same machine, communicating via HTTP/JSON over loopback. The setup measures the realistic HTTP framing, base64 serialisation of CKKS ciphertexts, FastAPI/uvicorn stack and JSON (de-)serialisation, but without physical network latency. The encoder is multilingual-e5-small ($d = 384$, $k = 192$, $M = 48$, $K_{\text{cands}} = 40$, $N_{\text{docs}} = 10^6$), averaged over $500 \text{ queries} \times 5 \text{ rotation seeds}$.

On-wire payload sizes. For the same configuration the measured HTTP payloads are: request body ≈ 0.29 MB (one CKKS query ciphertext, base64-encoded JSON, inflated $\approx 1.37\times$ over the 0.21 MB raw ciphertext); response body ≈ 11.4 MB ($K_{\text{cands}} = 40$ score ciphertexts, base64 JSON). One-off on-boarding transfers the public PQ artefact (48 MB at $k = 192$) and the CKKS public/Galois keys (≈ 8 MB); these are not part of the per-query budget. Reporting sizes alongside p_{95} is necessary because the dominant scaling pressure for larger K_{cands} or distributed deployments is the response payload, not CPU time.

Scalability to large corpora. The cryptographic cost is independent of the corpus size N_{docs} . The server reranks only the K_{cands} -document short-list returned by stage-1 PQ filtering, so the ciphertext-plaintext budget is $O(K_{\text{cands}} \cdot k)$: the 283 ms of Table 7 is the cost at $K_{\text{cands}} = 40$,

Table 7: Latency decomposition at the canonical operating point ($k = 192$) on a 10^6 -document corpus. The k -dependent server-side ct-pt rerank is measured on the 5-seed integral run; the k -independent client-side and HTTP stages are measured on the client-server HTTP-loopback PoC. The encoder forward pass is independent of the defence layer and excluded from the total.

Stage	p_{95} (ms)	Notes
Encoder forward pass (client)	~ 200	encoder-dependent; excluded from total
Projection to $\text{span}(V_k)$ (client)	0.1	negligible
Local PQ search (client, faiss)	22.8	$M = 48$, $K_{\text{cands}} = 40$, 10^6 docs
CKKS encryption (client)	4.9	one rotated query, $N_{\text{poly}} = 8192$
Server CKKS rerank, ct-pt	283	40 ct-pt ops at $k = 192$ (5-seed integral run)
HTTP + serialisation (req+resp)	38	base64 ciphertexts over loopback
CKKS decrypt + sort (client)	20.7	40 score ciphertexts
Total (excluding encoder)	≈ 370 ms	p_{95} , full pipeline

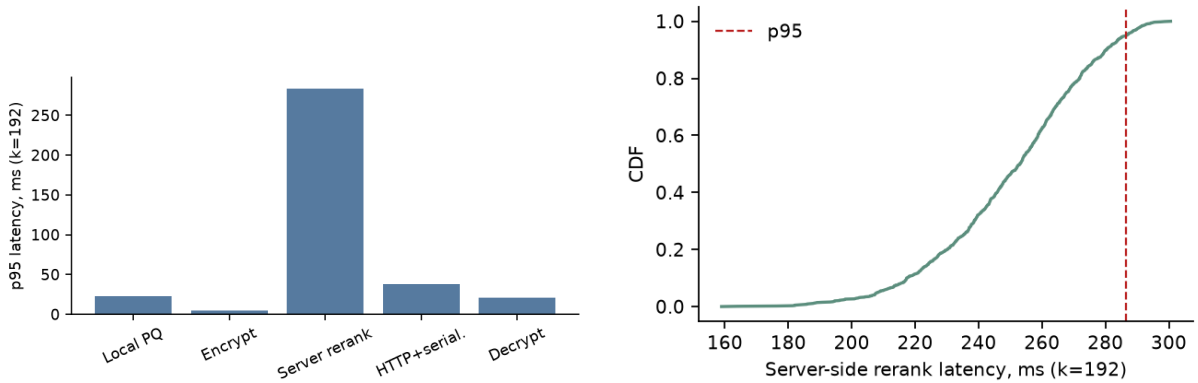


Figure 5: Left: per-stage latency decomposition at $k = 192$ (the server-side ct-pt rerank dominates). Right: CDF of the server-side rerank latency over 500 queries $\times$ 5 seeds ($p_{95} \approx 283$ ms); the full end-to-end $p_{95} \approx 370$ ms (Table 7) stays well below the 1 s SLA.

$k = 192$ and does *not* grow with N_{docs} . Only the local stage-1 PQ search is $O(N_{\text{docs}})$, and at 10^6 documents it already costs just 22.8 ms on a single faiss index (and is trivially shardable). Storage is linear and modest—the public PQ artefact is 48 MB at 10^6 documents and $k = 192$. The server stage is moreover embarrassingly parallel (independent ct-pt dot products across the short-list and across queries), so end-to-end throughput scales horizontally. The sub-second budget therefore holds at 10^6 documents and is preserved as the corpus grows: the only quantity scaling with N_{docs} is the cheap plaintext PQ stage, not the CKKS reranking.

7.5 Experiment 5: semantic-DLP detection in the protected space

The detection-fidelity Corollary 1 predicts that leak detection carried out in the protected, SVD-truncated space tracks a plaintext detector up to a score distortion of at most $2\sigma_{\text{rec}}^2$. We test this directly, on the same 10^6 -document Russian-Wikipedia corpus and the same five encoders as Experiment 7.3.

Canary protocol. We treat 250 documents as a confidential reference set and embed them, together with 50,000 unrelated background documents, into a protected index of size $M = 50,250$. The projection V_k ($k = d/2$) is the top- k right-singular subspace of the corpus—exactly the operating point whose σ_{rec} is reported in Experiment 7.3. As candidates we use 500 short canaries, each the leading sentence of a known source document and hence a genuine textual fragment of it. A canary is a true leak when its source document is one of the 250 secrets (250

Table 8: Semantic-DLP detection in the protected (SVD-truncated, $k = d/2$) space versus plaintext, on a 250/250 canary split against a reference corpus of $M = 50,250$ (250 secret documents plus 50,000 background). Detection AUC is at near parity (worst gap -0.013), and the Corollary 1 bound $|\hat{s} - s| \leq \varepsilon(\sigma_c, \sigma_r)$ is never violated across 1.5×10^6 sampled pairs. The measured σ_{rec} reproduces the values of Experiment 7.3 to ± 0.001 .

Encoder	d	σ_{rec}	AUC _{plain}	AUC _{prot}	Δ AUC	$\max \hat{s} - s $	viol./1.5M
E5-SMALL	384	0.238	0.914	0.901	-0.013	0.107	0
E5-BASE	768	0.130	0.936	0.939	$+0.003$	0.070	0
MPNET	768	0.108	0.841	0.841	-0.001	0.015	0
E5-LARGE	1024	0.101	0.923	0.925	$+0.002$	0.042	0
BGE-M3	1024	0.129	0.929	0.929	0.000	0.021	0

positives) and a non-leak otherwise (250 negatives); the detector flags a candidate c whenever $\max_{r \in \mathcal{R}} \cos(c, r) \geq \tau$. Because the secret rotation cancels in the cosine, the protected score is the cosine of the two projections, so this experiment isolates the effect of the *truncation* on detection—the confidentiality of the rotation is treated separately in Section 8.

Results. Table 8 reports the detection AUC in the plaintext and protected spaces. Three observations follow. First, the measured σ_{rec} reproduces the values of Experiment 7.3 to ± 0.001 , confirming an identical pipeline. Second, detection in the protected space is at *near parity* with plaintext: the AUC gap stays within $[-0.013, +0.003]$ across all five encoders—well inside the 0.05 margin below which one would have to report a tunable trade-off instead—and for the two higher-dimensional retrieval encoders (E5-BASE, E5-LARGE) the projection slightly *improves* detection, the linear-denoiser effect anticipated after Corollary 1. Third, the bound itself is never violated: across 1.5×10^6 randomly sampled candidate–reference pairs (300,000 per encoder) the empirical distortion $|\hat{s} - s|$ falls below the per-pair bound $\varepsilon(\sigma_c, \sigma_r)$ in every case, the largest observed distortion being 0.107 (E5-SMALL, whose σ_{rec} is largest).

Generality across corpora, domains and languages. The 95% confidence intervals confirm the parity on the run above directly: every protected–plaintext gap in Table 8 is far inside the AUC confidence interval (Hanley–McNeil half-widths 0.02–0.035 at $n = 250+250$), so the two detectors are statistically indistinguishable. To show this is not an artefact of a single corpus, we repeat the test on five further collections spanning three languages and several domains—English BEIR (SCIFACT, NFCORPUS, ARGUANA) and multilingual MIRACL (Bengali, Swahili)—taking each query and its relevant passage as the canary pair (Table 9). Across all ten corpus–encoder combinations the protected detector again stays within the 95% bootstrap CI of the plaintext one ($|\Delta\text{AUC}| \leq 0.014$), and Corollary 1 is never violated: 0 exceptions in a further 2.0×10^6 pairs, i.e. 0 in 3.5×10^6 pairs over six corpora and a wide σ_{rec} range (0.09–0.23). Absolute detection difficulty varies with the task—near chance on ARGUANA counter-argument retrieval, high on MIRACL—which is orthogonal to, and does not affect, the protected-versus-plaintext parity.

Scope. This is a detection-*fidelity* result for an honest server, not a confidentiality guarantee: it shows that truncation preserves the ability to flag leaks, whereas the resistance of the reference set to reconstruction rests on the secret rotation and is quantified separately (Section 8). The canary is also a clean “fragment-in-the-clear” leak; robustness to deliberate paraphrase or obfuscation of the leaking text is a separate evaluation we leave to future work (Section 10).

7.6 Experiment 6: robustness of detection to leak obfuscation

Experiment 7.5 detects a verbatim fragment. A motivated insider will instead paraphrase or truncate the leaking text, so we re-run the protected detector against three obfuscations of each

Table 9: Corollary 1 and DLP detection across additional corpora (English BEIR: SCIFACT/NFCORPUS/ARGUANA; multilingual MIRACL: Bengali/Swahili), with E5-SMALL (S) and E5-BASE (B). $AUC \pm$ is the 95% bootstrap CI half-width (10^3 resamples); the canary is a query versus its relevant passage. Across every corpus the protected detector tracks plaintext within CI ($|\Delta| \leq 0.014$), and the bound is never violated. Detection difficulty itself varies by task (chance-level on ARGUANA counter-argument retrieval, high on MIRACL), which is orthogonal to the parity claim.

Corpus	Lang/domain	Enc	σ_{rec}	AUC_{plain}	AUC_{prot}	ΔAUC
SCIFACT	en/science	S	0.185	0.587 ± 0.064	0.589 ± 0.065	+0.002
NFCORPUS	en/medical	S	0.173	0.555 ± 0.060	0.570 ± 0.062	+0.014
ARGUANA	en/argument	S	0.201	0.511 ± 0.030	0.512 ± 0.030	+0.001
MIRACL-BN	bn/wiki	S	0.228	0.776 ± 0.043	0.778 ± 0.042	+0.001
MIRACL-SW	sw/wiki	S	0.218	0.651 ± 0.048	0.655 ± 0.049	+0.004
SCIFACT	en/science	B	0.092	0.582 ± 0.067	0.581 ± 0.067	-0.001
NFCORPUS	en/medical	B	0.086	0.557 ± 0.061	0.553 ± 0.060	-0.004
ARGUANA	en/argument	B	0.106	0.496 ± 0.030	0.493 ± 0.029	-0.003
MIRACL-BN	bn/wiki	B	0.124	0.791 ± 0.042	0.788 ± 0.043	-0.003
MIRACL-SW	sw/wiki	B	0.115	0.672 ± 0.044	0.674 ± 0.044	+0.002

Table 10: Robustness of protected-space DLP detection (AUC) when the leaking text is obfuscated: V0 verbatim leading sentence (Experiment 7.5), V1 word-order shuffle, V2 a 40% random word deletion, V3 a different (second) sentence of the same document. Negatives are benign non-member candidates; 250 secret leaks (195 for V3, the secrets whose source document has a usable second sentence).

Encoder	d	V0 verbatim	V1 shuffle	V2 delete-40%	V3 2nd-sent.
E5-SMALL	384	0.901	0.919	0.829	0.837
E5-BASE	768	0.939	0.941	0.883	0.891
MPNET	768	0.841	0.881	0.732	0.780
E5-LARGE	1024	0.925	0.914	0.852	0.901
BGE-M3	1024	0.929	0.867	0.757	0.892

secret’s leading sentence—a word-order shuffle (V1), a 40% random word deletion (V2), and substitution by a different (second) sentence of the same document (V3)—re-embedding every variant with each encoder; a sanity check confirms the re-embedding reproduces the cached vectors (mean cosine 1.000). Table 10 reports the result. Word-order shuffling is a weak evasion: the AUC change is small (between -0.06 and $+0.04$), because the short-fragment embeddings are largely order-insensitive. Aggressive deletion is the most effective attack—AUC falls by 0.06 to 0.17, worst for BGE-M3—yet detection stays well above chance (0.73 to 0.88), and substituting an entirely different sentence of the same document is still detected at 0.78 to 0.90. The 95% confidence intervals on these AUCs are ± 0.02 to ± 0.035 ($n = 250$ leaks versus 250 benign candidates), so the V2 and V3 reductions are significant whereas the V1 change sits within noise. Obfuscation therefore degrades but does not defeat protected-space detection; the residual gap under heavy deletion is the empirical motivation for a semantic-canary augmentation (Section 10).

Reference-corpus confidentiality and membership. Two further DLP-specific risks are already settled by the analyses elsewhere in this paper. First, a similarity-threshold detector is inevitably a *membership oracle*: probing it with a member document returns the maximal score

Table 11: Known-plaintext Procrustes attack against the secret rotation (m leaked pairs, means over five seeds). With $\approx k$ pairs the orientation is numerically recovered.

Known pairs m	Mean cosine	Rel. L_2 error	Target R@1	Target R@10
0	-0.004	1.417	0.000	0.001
10	0.098	1.343	0.005	0.039
50	0.377	1.116	0.722	0.925
100	0.621	0.870	1.000	1.000
192	1.000	1.8×10^{-5}	1.000	1.000
500	1.000	4.9×10^{-7}	1.000	1.000

(1.000) in both the plaintext and the protected space, because truncation preserves self-identity ($\pi_k(r)$ still matches $\pi_k(r)$); a deployment must therefore rate-limit queries or withhold raw scores, a property inherent to any matching-based DLP rather than to this construction. Second, the confidentiality of the reference set against an attacker who can align the protected geometry to plaintext is exactly the known-plaintext analysis of Section 8: the rotation is recovered and the reference paragraphs re-identified once the attacker holds on the order of k leaked pairs (Tables 11 and 12), so corpus confidentiality scales with k and with limiting plaintext-pair exposure. The ciphertext-plaintext check that would gate inline enforcement adds only one masked dot product per short-list entry and stays within the sub-second budget of Experiment 7.4. Fully learned cross-space inverters (VEC2VEC, ZERO2TEXT) require training a translator we do not run here and remain, as in Section 10, the main residual gap.

8 Security Analysis Under Known-Plaintext and Stronger Attackers

The experiments above establish the construction’s behaviour *inside* its non-adaptive threat model (Section 3). A privacy paper that stopped there would be open to the objection that the only tested adversary is also the weakest one. We therefore deliberately step outside the model and measure four stronger channels—known-plaintext recovery of the rotation, a text-level reference-corpus lookup, leakage from the public PQ artefact, and an aligned off-the-shelf inverter—followed by a baseline suite and a benchmark-generalty check. Several of these are *negative* results for broad document-privacy claims; reporting them is what licenses the deliberately narrow framing of Sections 9–11.

8.1 Known-plaintext recovery of the secret rotation

The secret rotation R is the only document-side layer beyond SVD truncation. We test it against a known-plaintext attacker who holds m pairs $(E(x_i), E_{\text{rot},i})$ and estimates the orientation with the orthogonal Procrustes solution [27]. The setup uses `e5-small` projected to $k = 192$, five rotation seeds, 400 held-out probes and a 10 000-document gallery; the attack is scored in embedding space, where a successful alignment lets a probe recover its own projected vector and retrieve the matching gallery document. This is the relevant precursor to a few-shot alignment attack because it hands the inverter the native projected space.

The result is unambiguous (Table 11, Fig. 6): the rotation is useful only while aligned plaintext anchors are unavailable. Even 50 pairs—well below the retained dimension $k = 192$ —already retrieve the target in the top 10 for 92.5% of probes in a 10 000-document gallery, and at $m \approx k$ the orientation is numerically exact. The rotation therefore raises the bar against an attacker with *no* aligned plaintext, but it is not a defence against known plaintext; this is reflected directly in the threat-coverage table.

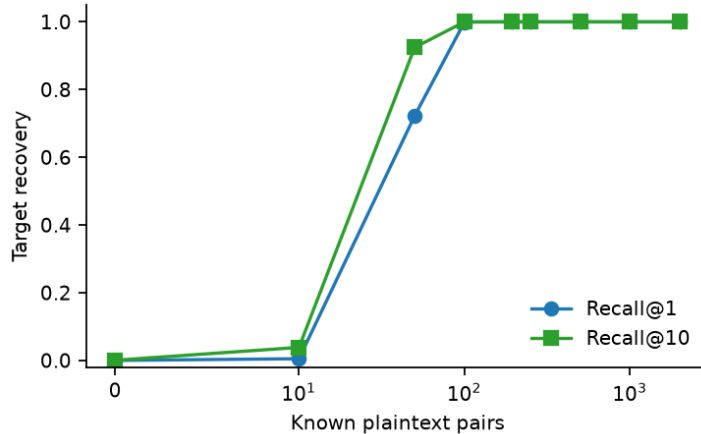


Figure 6: Known-plaintext alignment destroys the unknown-rotation assumption: with about k leaked pairs the rotation is recovered. The x axis uses a symlog scale to include $m = 0$.

Table 12: Reference-corpus lookup after known-plaintext alignment (overlapping reference: 500 targets + 100 000 decoys; Jaccard is token-set overlap with the top-1 retrieved paragraph; means over five seeds).

Known pairs m	Exact R@1	Exact R@10	Jaccard overlap-ref	Jaccard disjoint-ref
0	0.000	0.000	0.006	0.006
10	0.000	0.009	0.009	0.009
25	0.044	0.132	0.055	0.012
50	0.535	0.784	0.543	0.018
100	0.998	1.000	0.999	0.034
192	1.000	1.000	1.000	0.043

8.2 Reference-corpus lookup after alignment

The Procrustes test is in embedding space. We next ask whether the same failure becomes a *text-level* channel without training any generative inverter. The attacker is given the protected vectors, m known pairs for estimating R , and a reference corpus of candidate paragraphs with native `e5-small` embeddings—modelling the realistic case where some private documents also appear in a public or previously leaked corpus. We use 500 targets, 100 000 decoys, five seeds and the same $k = 192$ subspace; the overlapping reference contains the targets, the disjoint reference removes them and serves only as a lexical nearest-neighbour proxy.

The overlap case is severe (Table 12): with only 50 known pairs the attacker recovers the exact source paragraph at 53.5% top-1 among 100 500 candidates, rising to 99.8% at $m = 100$. This is the text-level analogue of the Procrustes result and makes the static-side limitation concrete: if a public or leaked reference corpus overlaps the private collection, rotation secrecy is insufficient. The disjoint-reference column is reported as a boundary condition—token Jaccard stays at 0.034 ($m = 100$), so a non-overlapping reference needs semantic metrics or a generative inverter before text reconstruction can be claimed.

8.3 Leakage from the public PQ artefact

Stage-1 candidate filtering publishes a product-quantization codebook plus per-document codes, trained in the rotated space. We quantify how much of E_{rot} this public artefact alone reconstructs on a 20 000-document `e5-small` sample (Table 13). The canonical configuration ($M = 48$, 8 bits) preserves mean cosine 0.953 to E_{rot} , 67.4% of exact top-10 neighbours, and 97.1% of exact top-10 neighbours inside the approximate top-40 candidate set—that is, the public codes alone almost perfectly reproduce the candidate set the server reranks.

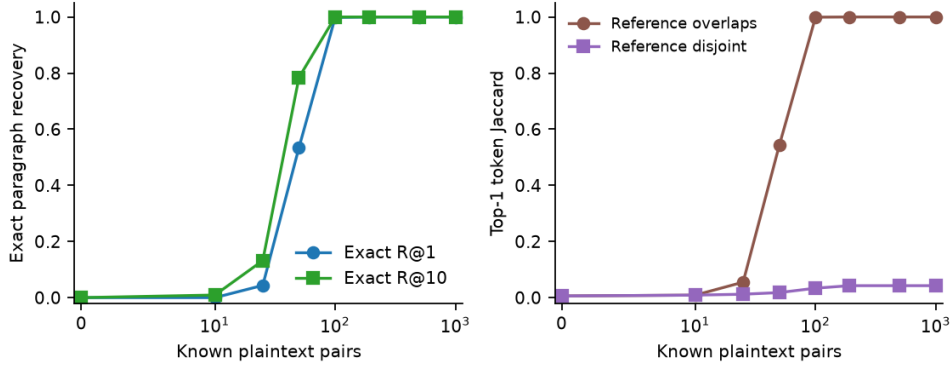


Figure 7: Reference-corpus lookup after alignment. When the reference overlaps the protected collection, alignment turns the protected vector into an exact paragraph lookup; with a disjoint reference this token-overlap proxy stays low.

Table 13: Leakage from the public PQ codes (rotated `e5-small`, 20 000 documents; neighbour overlap excludes the query document).

M	Bits	Bytes/vector	Cosine to E_{rot}	NN overlap@10
24	8	24	0.837	0.384
48	8	48	0.953	0.674
48	6	36	0.904	0.529

These numbers do not weaken the CKKS query-privacy contribution, but they do narrow the document-privacy interpretation: both the known-plaintext anchor channel and the public PQ artefact must be treated as exposed. A deployment that needs stronger static-side confidentiality requires either a different static primitive or a composition that hides the PQ artefact itself.

8.4 Aligned off-the-shelf inversion stress test

Finally we run a generative inverter against the aligned space. The attacker is given GTR-base embeddings ($d = 768$), SVD truncation to $k = d/2$, the secret-rotation setting, and up to $m = 500$ known pairs for Procrustes alignment; the inverter is the off-the-shelf VEC2TEXT corrector with a larger budget than Experiment 7.2 (24 corrector iterations, max input length 96). For deterministic reproducibility on a host that could not fetch external corpora the run uses the script’s offline synthetic-news+PII corpus (120 held-out texts). The decisive control is the raw-embedding row.

The reading is honest but limited (Table 14): no exact document or typed PII is recovered even by a known- R oracle, yet the *raw* embedding row is also at the BLEU floor (0.013), so

Table 14: Aligned off-the-shelf VEC2TEXT stress test (120 held-out texts; m is the number of pairs used for alignment). Exact match and typed-PII recall are zero in every row—but so is the raw control, so this corrector cannot *certify* protection.

Case	Token F1	BLEU	Exact	PII recall
Raw embedding	0.133	0.0133	0.000	0.000
Known- R oracle	0.138	0.0134	0.000	0.000
Unknown R (= Procrustes $m=0$)	0.121	0.0129	0.000	0.000
Procrustes, $m = 50$	0.126	0.0126	0.000	0.000
Procrustes, $m = 100$	0.141	0.0127	0.000	0.000
Procrustes, $m = 192$	0.134	0.0132	0.000	0.000
Procrustes, $m = 500$	0.135	0.0133	0.000	0.000

Table 15: Acc@1 under lightweight projection baselines at $k = d/2$ (random baselines: mean $\pm$ s.d. over three seeds).

Encoder	Raw	SVD	Gaussian RP	Orthogonal RP
e5-small	0.782	0.702	0.688 ± 0.016	0.737 ± 0.010
e5-base	0.830	0.834	0.781 ± 0.011	0.804 ± 0.017
mpnet	0.716	0.650	0.698 ± 0.003	0.705 ± 0.008
e5-large	0.796	0.814	0.765 ± 0.010	0.769 ± 0.009
bge-m3	0.818	0.832	0.789 ± 0.004	0.810 ± 0.002

Table 16: SVD truncation vs. independent Gaussian noise at matched mean relative distortion (100 000-document gallery). NN overlap@10 is the overlap with raw-space neighbours; lower means less raw geometry preserved.

Encoder	k/d	σ_{rec}	SVD Acc@1	SVD NN@10	Noise Acc@1	Noise NN@10
e5-small	0.125	0.385	0.318	0.232	0.750	0.283
e5-small	0.250	0.332	0.600	0.333	0.788	0.367
e5-small	0.500	0.237	0.814	0.468	0.828	0.531
e5-small	0.875	0.079	0.880	0.551	0.872	0.830
e5-base	0.125	0.354	0.708	0.370	0.864	0.503
e5-base	0.250	0.268	0.872	0.501	0.886	0.641
e5-base	0.500	0.129	0.916	0.578	0.902	0.829
e5-base	0.875	0.023	0.920	0.592	0.910	0.969

this off-the-shelf corrector is too weak to certify the transform. It does not contradict the exact-lookup failure of Section 8.2; the remaining open risk is a learned or corpus-adapted decoder trained end-to-end on the rotated space, which we keep as the highest-priority future experiment (Section 10).

8.5 Baseline suite: projection and calibrated noise

To position the transform against natural alternatives rather than against the raw system alone, we compare SVD projection with Gaussian random projection and random orthogonal projection [43] at the same $k = d/2$ (Table 15), and with calibrated isotropic Gaussian noise matched at equal mean relative distortion (Table 16).

Two engineering conclusions follow. First, the projection choice is encoder-dependent: SVD is strongest on the retrieval-trained $d \geq 768$ encoders that are the method’s operating region, but random orthogonal projection beats SVD on `mpnet` and `e5-small`—so a deployment should run a baseline sweep, not assume SVD is universally best. Second, σ_{rec} is not a privacy metric: at $k/d = 0.875$ matched noise preserves almost all raw neighbours, and at the main $k/d = 0.5$

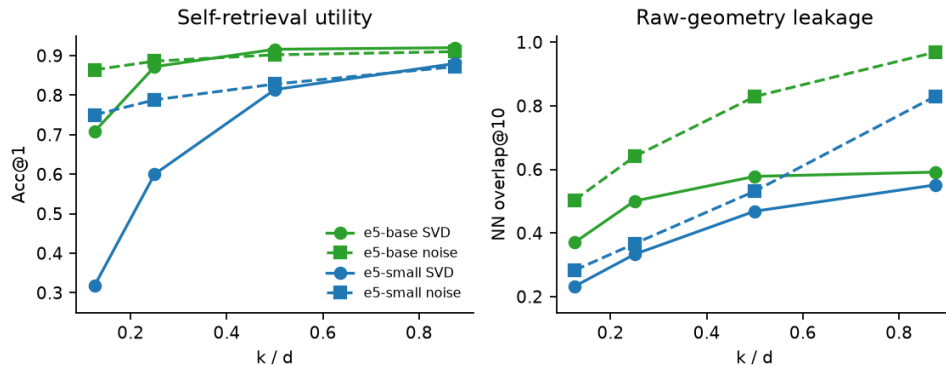


Figure 8: Utility/leakage diagnostic for SVD truncation and calibrated Gaussian noise (dashed: matched- σ_{rec} noise baselines).

Table 17: SVD truncation ($k = d/2$) vs. the raw space on BEIR. $\Delta = \text{SVD} - \text{raw}$; brackets are paired-bootstrap 95% CIs (10^4 resamples). **Every** Δ nDCG@10 CI excludes 0.

Encoder	Dataset	σ_{rec}	raw nDCG@10	SVD nDCG@10	Δ nDCG@10 (95% CI)
e5-large	SciFact	0.072	0.643	0.574	-0.069 [-0.091, -0.047]
e5-base	SciFact	0.090	0.637	0.586	-0.051 [-0.072, -0.032]
e5-small	SciFact	0.183	0.598	0.519	-0.079 [-0.103, -0.055]
e5-base	NFCorpus	0.085	0.327	0.305	-0.022 [-0.032, -0.013]
e5-small	NFCorpus	0.172	0.302	0.277	-0.025 [-0.035, -0.014]

operating point SVD removes far more raw-neighbour structure than matched noise (0.578 vs. 0.829 NN overlap@10 on **e5-base**) while keeping Acc@1. The defensible claim is thus the narrow one: SVD is an encoder-dependent utility/leakage trade-off, not a noise mechanism that dominates all alternatives.

8.6 Benchmark generality on BEIR

The integral experiment uses one-million-document self-retrieval, a controlled geometry probe. To check that the metric-preservation reading is not an artefact of self-retrieval, we repeat the raw-vs-SVD comparison on two standard BEIR [44] datasets with graded relevance judgements—SciFact (5,183 docs, 300 queries) and NFCorpus (3,633 docs, 323 queries)—fitting V_k on the corpus at $k = d/2$ and comparing exact inner-product retrieval in $\mathbb{R}^d$ and in $\text{span}(V_k)$.

Table 17 is a deliberate corrective to the self-retrieval reading. On both graded-relevance datasets and for every encoder, half-SVD truncation *significantly reduces* nDCG@10 (all CIs exclude zero, drops of 2–8 points). Decisively, **e5-large**—the one encoder that *gained* on the 10^6 self-retrieval probe—*loses* 0.069 nDCG@10 on SciFact. The denoiser effect is therefore real but *task-dependent*: on real graded IR the discarded spectrum still carries relevance, so k must be treated as a tunable accuracy cost via a k -sweep, exactly as the encoder-selection discussion below recommends. (Full end-to-end CKKS reranking on BEIR is left to future work; this experiment isolates the projection-fidelity property that the reranking preserves.)

9 Discussion

Encoder selection is part of the protocol. The protective wrapper is essentially metric-preserving in the projected space; the end-to-end deviation from the raw baseline is dominated by what SVD truncation does on the chosen encoder. Models trained with contrastive retrieval objectives (E5-*, BGE-*) concentrate the retrieval signal in the top singular directions and recover most of the original quality after $k = d/2$ truncation; some of them even *benefit* from the truncation (linear-denoiser effect). Paraphrase-distilled models (MPNET) and very compact models (E5-SMALL) do not show this concentration, and we recommend either replacing them with a retrieval-trained equivalent or running a k -sweep before deployment to find a k that simultaneously satisfies the application’s accuracy budget and $\sigma_{rec} \geq 0.10$.

The role of the rotation R : obfuscation, not a primitive. The secret rotation is an empirical obfuscation layer, not a cryptographic primitive. It contributes a defence channel orthogonal to SVD truncation, and even at $k/d = 1.0$ the unknown- R regime reduces off-the-shelf Vec2Text BLEU to the noise floor. Its limits are now measured rather than asserted (Section 8): a known- R attacker gains nothing beyond SVD; a known-plaintext attacker recovers R by orthogonal Procrustes from $\approx k$ leaked pairs (Table 11), which becomes an exact paragraph lookup when a reference corpus overlaps the collection (Table 12); and an end-to-end learned decoder remains the main untested case. The rotation’s value is therefore that it is the cheapest architectural layer with a measurable effect on an attacker who lacks aligned plaintext—not a security guarantee. The

contribution of this paper is the measured privacy/latency/accuracy trade-off and its quantified boundaries, not the rotation in isolation.

What CKKS hides and what it does not. The CKKS layer hides the numerical representation of the query and the exact similarity scores from the server. Two things it does *not* hide are the identifiers of the candidates produced by stage-1 PQ filtering and the L_2 -distances among them implicit in the order in which they are sent. An access-pattern attacker can therefore, over many queries, build a co-occurrence model of which documents are co-retrieved. Composing the pipeline with a PIR-style retrieval primitive (Tiptoe [34], SealPIR [35], OnionPIR [37], SimplePIR [36]) is the natural remedy; ours is engineered to be PIR-friendly because the ct-pt operations on the server are stateless and cluster-local.

Parameter selection vs. hand-tuning. Compared with hand-tuned TenSEAL stock parameters, the configuration chosen by the offline grid search (Section 5) achieves a $\approx 1.7\times$ speed-up at the same security level and the same ranking quality (within $\Delta\text{Acc}@1 \leq 1$ p.p. of the SVD-projected baseline). The selection is a one-off offline cost and adds nothing at run time; porting to a different hardware platform only requires re-running the same small benchmark grid.

The construction as a leak-detection primitive. The same asymmetric geometry serves as a privacy-preserving semantic data-loss-prevention index for an LLM firewall: the protected static side becomes a confidential “must-not-leak” reference corpus, and a candidate prompt, output or retrieved passage is screened against it without the server holding readable embeddings. Two results make this concrete. Detection in the protected space tracks a plaintext detector to within 0.013 AUC across all five encoders (Experiment 7.5), the gap provably confined to a $2\sigma_{rec}^2$ score band (Corollary 1)—and, as in the retrieval setting, the truncation *helps* the strongest encoders, the same linear-denoiser effect. The detector degrades but does not break under obfuscation of the leaking text (Experiment 7.6). The honest caveats carry over: this is detection *fidelity*, not corpus confidentiality—the reference set inherits the same known-plaintext recovery limit as the document side (Section 8)—and the detector is inherently a membership oracle, so a deployment must withhold raw scores.

10 Limitations and Required Experiments

We separate *intrinsic limitations* (properties of the construction under study) from the *remaining open experiments*. The stronger-attacker, baseline and generality experiments that earlier drafts listed as mandatory future work are now reported in Section 8; what is listed below as open is the genuinely residual set, kept as named scoped experiments rather than vague “future work” as a deliberate hedge against overclaiming.

Intrinsic limitations.

- *Document privacy is not cryptographic.* E_{rot} is plaintext on the server; it is protected only by SVD truncation (a proxy) and the secret rotation (an empirical obfuscation layer). A known- R , known-plaintext or adaptive adversary defeats this layer.
- *Access patterns and PQ codes are exposed.* CKKS hides values, not which candidates are reranked, and the public PQ codes are a lossy view of E_{rot} . Production use requires composition with a PIR/ORAM layer plus a PQ-leakage analysis.
- *Encoder-dependence of the accuracy budget.* The strict end-to-end criterion $\Delta\text{Acc}@1 \geq -5$ p.p. holds only on retrieval-trained encoders with $d \geq 768$. Compact or paraphrase-distilled encoders need either replacement or a less aggressive k (which then violates the $\sigma_{rec} \geq 0.10$ proxy).

- *Single-machine evaluation.* The latency numbers in Experiment 7.4 are taken on loopback HTTP; a geographically distributed deployment will add (uncharacterised) network latency.

Completed in this revision. The stronger-attacker programme is now executed (Section 8): known-plaintext Procrustes recovery of R (Table 11); the text-level reference-corpus lookup (Table 12); public-PQ-code leakage (Table 13); an aligned off-the-shelf inverter (Table 14); a projection and calibrated-noise baseline suite covering random projection, SVD-without-rotation and matched-distortion Gaussian noise (Tables 15–16); and a BEIR generality check on the projection step (Table 17).

Remaining open experiments.

1. *End-to-end learned decoder (highest priority).* The aligned off-the-shelf corrector of Section 8.4 is too weak to *certify* protection—its raw control is already at the BLEU floor. The decisive test is an inversion model *trained* on rotated/SVD embeddings of part of the corpus and evaluated on a holdout; the current narrative explicitly does not claim robustness against it. This trained-decoder evaluation against the rotated/SVD geometry—together with encoder-level adversarial defences that resist it—is the subject of dedicated follow-up work.
2. *Typed-PII, multi-encoder σ_{rec} ablation.* Replace the single-encoder, BLEU-only calibration of the $\sigma_{rec} = 0.10$ proxy with a multi-encoder ablation reporting BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms), which is what justifies (or refutes) the threshold.
3. *Modern attack suite and full-pipeline benchmarks.* Repeat the inversion study against the 2024–2026 attack generations that exploit the universal geometry of embedding spaces rather than a fixed basis orientation—few-shot alignment (ALGEN [10]), zero-shot inversion (ZSINVERT [11]), unsupervised cross-space translation (VEC2VEC [12]) and online cross-domain inversion (ZERO2TEXT [13])—with real text, and run the *complete* CKKS reranking (not only the projection step measured in Table 17) on BEIR, MIRACL and MS MARCO.
4. *Hardening the DLP reading.* Experiments 7.5 and 7.6 show protected-space detection at near parity and its graceful degradation under text obfuscation; the open work is a *semantic-canary* augmentation that catches the paraphrased leaks the 40%-deletion attack still erodes, a membership-inference hardening that rate-limits or perturbs the returned scores, and a leakage study of the reference corpus under the learned cross-space inverters of item 3.

11 Conclusion

This paper is a scientific study of how SVD truncation and an unknown secret rotation reshape the privacy–accuracy–latency trade-off of semantic search; it is not a systems-engineering report and does not claim a deployable system. Four findings answer the research questions. **(RQ1)** A formal lower- L_2 bound (Lemma 1) characterises exactly what a projection-restricted decoder cannot recover; it is a projection bound, not an inversion-security theorem, and it makes the *proxy* criterion $\sigma_{rec} \geq 0.10$ precise only within that decoder class, the threshold itself being an engineering proxy. **(RQ2)** At 10^6 -document scale the construction’s protective layer is metric-preserving in $\text{span}(V_k)$ (within the 5-seed CI), the query side is cryptographically protected by CKKS at sub-second p_{95} , while document and access-pattern privacy are *not* cryptographic and the public PQ artefact is an exposed channel whose leakage we now quantify (Section 8)—so the secret rotation is characterised as an empirical obfuscation effect against an off-the-shelf inverter, not as a guarantee. **(RQ3)** For retrieval-trained encoders with $d \geq 768$, data-driven SVD truncation to $k = d/2$ acts as a linear *denoiser*, improving the four ranking metrics over the raw baseline rather than degrading them—a finding of independent interest beyond the privacy

setting. **(RQ4)** The same asymmetric geometry doubles as a privacy-preserving leak-detection primitive: a server storing only the truncated, rotated vectors flags whether a candidate matches a protected reference corpus at near parity with a plaintext detector (Experiment 7.5; AUC gap within 0.013), the gap provably confined to a $2\sigma_{rec}^2$ band (Corollary 1) and degrading gracefully under obfuscation (Experiment 7.6), while the reference corpus inherits the document side’s known-plaintext confidentiality limit.

We do not claim the privacy problem is solved. This work delivers a measured privacy, accuracy and latency operating point and a characterised denoising phenomenon under a clearly delimited threat model, together with a stronger-attacker analysis (Section 8) that quantifies where the document-side obfuscation fails—known-plaintext recovery of R , reference-corpus lookup and public-PQ-code leakage—and leaves an end-to-end learned decoder as the principal open question, which we pursue in dedicated follow-up work. Composing the protocol with a PIR-style access-pattern hider—natural given the stateless server-side ct-pt operations—is the most attractive architectural extension.

Reproducibility

All scripts, notebooks, configuration files and JSON output of the experiments are released at <https://github.com/sergkurilenko/research/tree/article1>. The integral experiment is driven by a single Python script with deterministic seeds and a JSON output file; the CKKS parameter-selection benchmark ships its configuration grid and timings as a CSV. The corpus (one-million Russian-Wikipedia paragraphs) is reproducibly sliced from a public Wikipedia dump with the date and content checksum recorded.

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